ECE60827 CUDA Assignment 4
Jerry Liu, 5/5/2023

Overview:

In this assignment, the various functions and parts from the previous two programming assignments were connected together to form the AlexNet CNN. In total, it consists of 11 operations that are performed on an initial 227x227, 3 channel input. The general approach used for this assignment was to generate and pass managed/unified input/output pointers between the different kernels, which should, in theory, reduce the total amount of memory transferred between the host and the device (since outputs are only used by kernels). A possible improvement to this scheme could have been to allocate a single device pointer for each input and output tensor, and then resize the TensorShape variables associated with the pointers as needed. This optimization would reduce the total number of memory allocation calls, which would save both time and space. Unfortunately, this was not implemented due to complexity and time constraints.

Test Results:

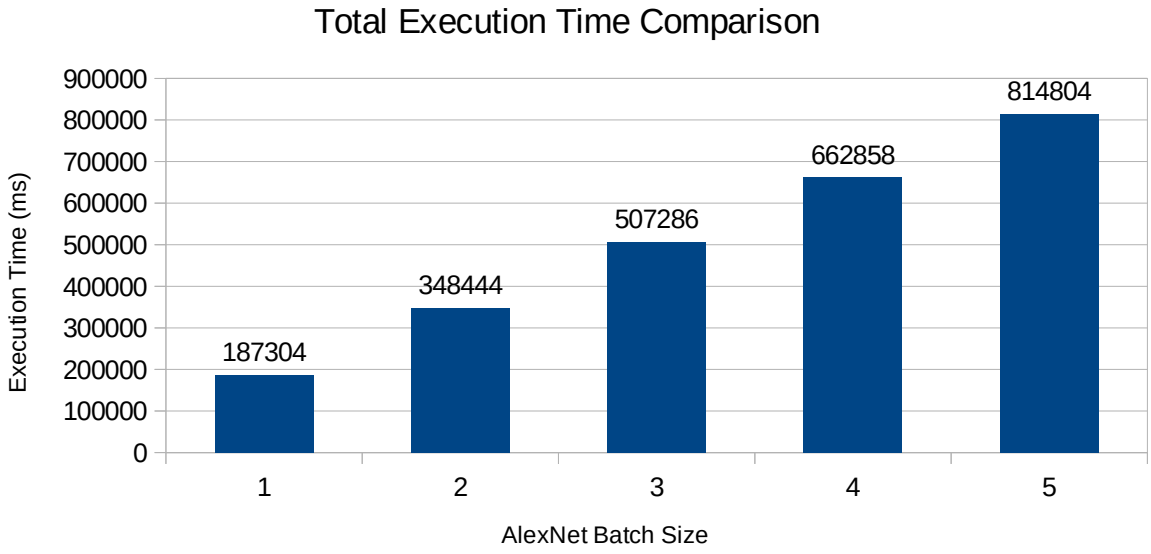

Figure 1: All execution times were measured using std:chrono.

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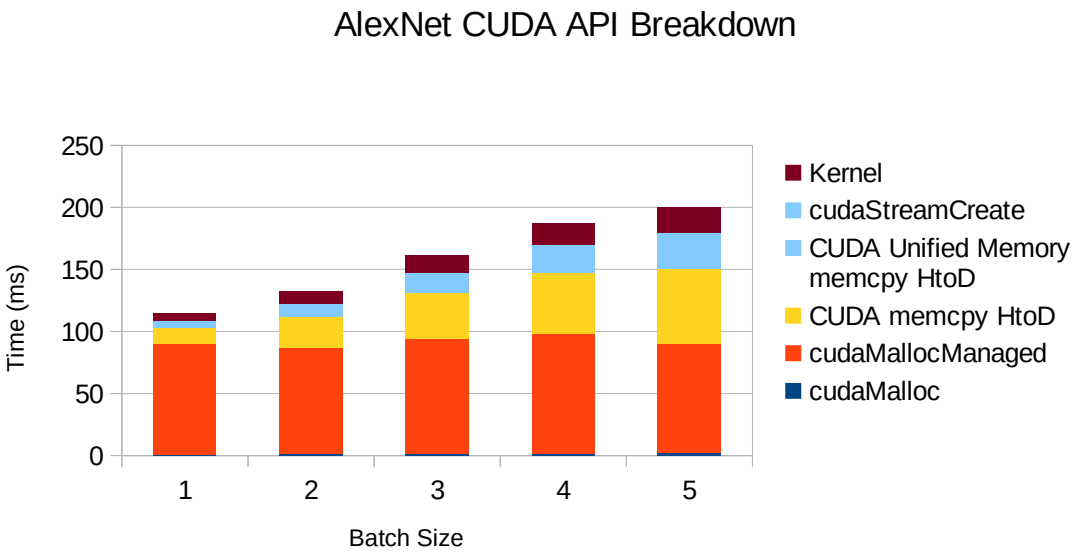

Figure 2: Breakdown of GPU convolution API calls using command "nsys nvprof ./lab3" Non-relevant API calls are excluded (i.e. cudaLibraryLoadData) for clarity.

Observations:

As expected, the execution time of the kernels grows as the batch size increases. Interestingly, while both the normal and unified cudaMemcpy's appear to also scale with batch size, the normal variant seems to scale faster than the unified variant. This is likely caused by the fact that the normal cudaMemcpy is used to copy the filters and weights to the device, of which obviously needs to scale with batch size (just like the input/output tensors). Unfortunately, there isn't a particularly easy method to reduce this overhead other than to create a specific kernel to generate these tensors on the device, which may not be possible depending on how these values are created. Regardless, based on the rate of growth of the various API calls, this implementation of AlexNet appears to be slightly memory bound based on the need for more tensor data to be copied to the device.